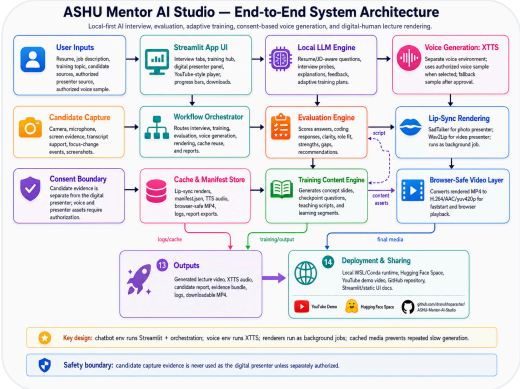
Self Project

2026-06-16

ASHU Mentor AI Studio: End-to-End AI Interview, Evaluation, Adaptive Training, and Digital-Human Lecture Platform



ASHU Mentor AI Studio is a local-first AI mentor platform that connects resume-aware interviews, JD-based evaluation, adaptive training, evidence capture, consent-based voice generation, and digital-human lecture rendering into one complete learning workflow.

Project Snapshot

DATE

2026-06-16

CATEGORY

Self Project

SHORT DESCRIPTION

ASHU Mentor AI Studio is a local-first AI mentor platform that connects resume-aware interviews, JD-based evaluation, adaptive training, evidence capture, consent-based voice generation, and digital-human lecture rendering into one complete learning workflow.

TAGS / TECH STACK

Agentic AI · Generative AI · Interview AI · Digital Human · Voice AI · Education AI · Training · Evidence Pack

Project Links

Demo Video | Hugging Face Space | GitHub Link | GitHub Link | GitHub Link

GitHub Link | GitHub Link | GitHub Link | GitHub Link | GitHub Link | Demo Video

GitHub Link | GitHub Link | GitHub Link | GitHub Link | GitHub Link | GitHub Link

GitHub Link | GitHub Link

Full Project Content

> **ASHU Mentor AI Studio** — **Adaptive Smart Human-like Unit Mentor AI Studio** — is an end-to-end AI mentor platform that moves beyond simple mock interviews. It reads a candidate profile, understands the target role, asks structured questions, evaluates responses, identifies learning gaps, recommends personalized training, and can convert training content into a digital-human lecture with consent-aware voice and lip-sync generation.

GitHub Repository: dranubhaparashar/ASHU-Mentor-AI-Studio

> Demo and live app: YouTube Demo · Hugging Face Space
>
> Wiki documentation: Home · Demo & Live App · Architecture & Product Design · Complete Technical Specification · Training & Interview Workflow
>
> Governance and evidence: Digital Presenter Voice & Lip-Sync Pipeline · Evidence Pack & Reporting · Security, Consent & Governance

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Demo Video

Embedded media: Open video

One-Line Idea

ASHU Mentor AI Studio turns a resume, job description, training topic, and candidate interaction into a complete AI mentoring cycle: interview → evaluation → learning gaps → adaptive training → digital-human lecture → evidence-backed report.

Most interview tools stop at question generation. ASHU Mentor AI Studio is designed as a complete workflow platform where assessment, training, generated media, and documentation remain connected.

Why This Project Exists

Interview preparation, candidate screening, and technical training are usually fragmented across multiple tools:

- one tool generates questions,
- another records candidate answers,
- another gives generic feedback,
- another prepares training material,
- and a completely separate pipeline is needed to create voice or video-based learning content.

Career preparation is fragmented across separate interview, resume, coaching, voice, and training tools. A system may ask questions, but it rarely converts answer quality into a meaningful training pathway. It may generate lessons, but the lessons are not always linked to the candidate's actual weaknesses. It may create videos, but the video generation workflow is often disconnected from consent, evidence, and evaluation.

ASHU Mentor AI Studio solves this by joining the full chain into one local-first AI mentor studio.

Project at a Glance

Area	Description
Full Form	Adaptive Smart Human-like Unit Mentor AI Studio
Primary Goal	Build an AI mentor that can conduct structured interviews, evaluate answers, recommend training, and generate digital-human teaching content.
Main Users	Students, job seekers, interview coaches, training teams, educators, HR reviewers, AI product demonstrators.
Core Interface	Streamlit dashboard with interview, evaluation, training, digital presenter, media, and report modules.
AI Layer	Local LLM orchestration for question generation, evaluation reasoning, training scripts, and feedback.
Media Layer	XTTS/Coqui voice generation, SadTalker/Wav2Lip lip-sync, FFmpeg browser-safe MP4 conversion.
Evidence Layer	Reports, transcripts, scores, logs, generated media manifest, and candidate evidence bundle.

Area	Description
Deployment Surface	GitHub repository, Hugging Face Space, YouTube demo, and GitHub Wiki documentation.
Design Principle	Candidate evidence and presenter assets are separated through explicit consent boundaries.

What Makes ASHU Different

> [!IMPORTANT] ASHU Mentor AI Studio is not just an LLM wrapper. It is a connected workflow where each output becomes useful for the next stage.

Ordinary Tool	ASHU Mentor AI Studio
Generates generic interview questions	Generates resume-aware and JD-aware questions
Gives basic feedback	Produces structured score, role-fit reasoning, and learning-gap analysis
Ends after mock interview	Converts weak areas into adaptive training modules
Produces static text	Can generate digital-human lecture output
Treats media generation separately	Connects training script, voice, lip-sync, cache, and final MP4 playback
Lacks evidence structure	Exports candidate reports, transcripts, logs, scores, and media manifests
Ignores consent boundaries	Separates candidate evidence, presenter source, and authorized voice assets

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Reader Walkthrough

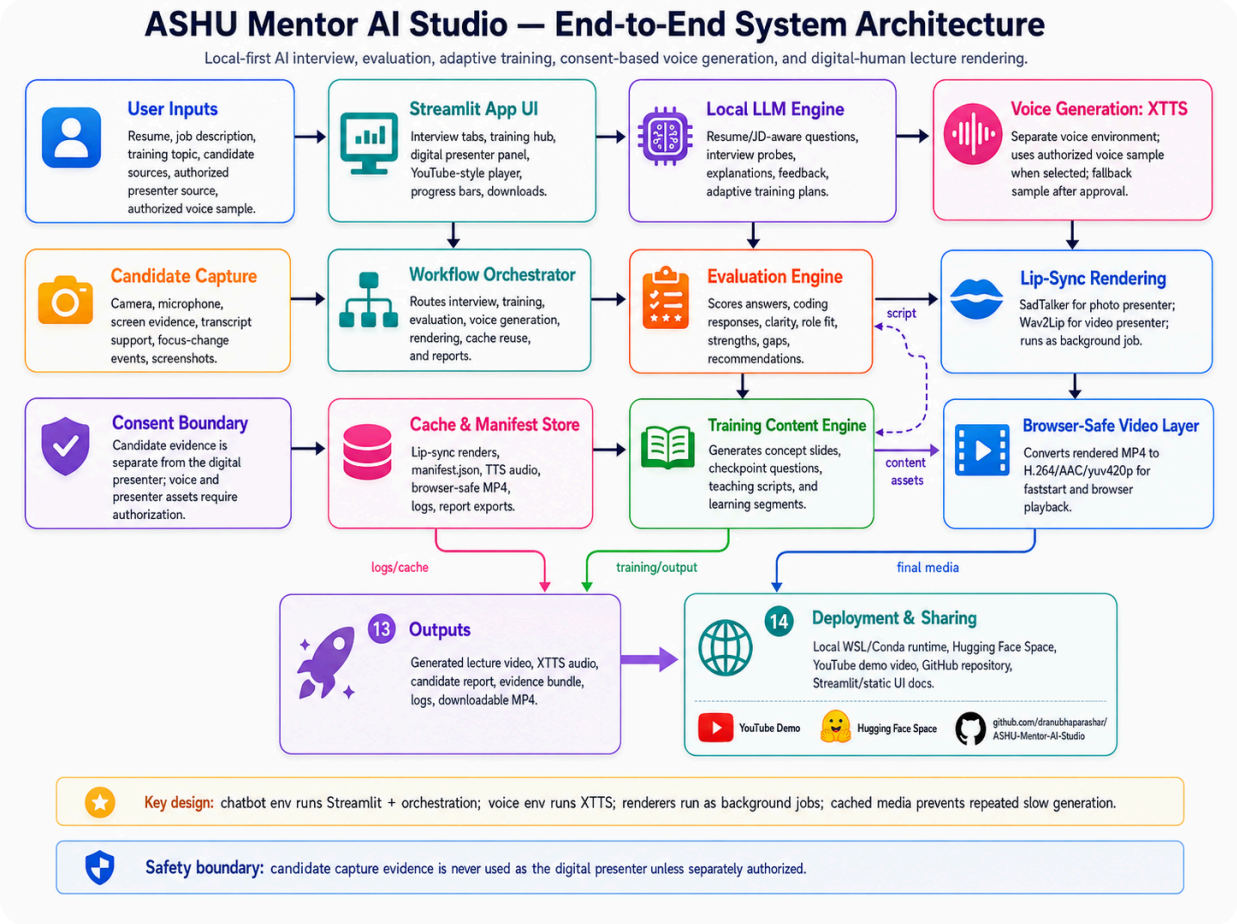
A reader reviewing this project can understand it in four steps:

1. **Input** — upload or paste a resume, job description, candidate details, and training topic.
2. **Interview** — the system generates structured role-aware questions and captures candidate responses.
3. **Evaluation** — answers are scored with strengths, weaknesses, role fit, readiness, and learning gaps.

4. **Training Output** — the platform converts gaps into learning material and can render a digital-human teaching video.

The result is not only a score. It is a complete learning and evidence package.

System Architecture



ASHU Mentor AI Studio follows a **local-first, consent-aware, multi-stage architecture**. The platform starts from user inputs such as a resume, job description, training topic, candidate answers, and optional presenter or voice assets. The Streamlit interface routes these inputs into the workflow orchestrator, which coordinates LLM reasoning, evaluation, training generation, media creation, caching, and report export.

The architecture has four major boundaries:

1. **Candidate boundary** — candidate responses, transcript, scores, and interview evidence are used for assessment and feedback.
2. **Presenter boundary** — presenter image or video is used only for digital-human rendering when explicitly authorized.
3. **Voice boundary** — XTTS voice generation requires an authorized voice sample or approved fallback voice.

4. **Deployment boundary** — heavy media rendering works best locally, while Hugging Face provides a public demo surface.

```
flowchart LR
    U[User Inputs] --> UI[Streamlit App UI]
    CAP[Candidate Capture] --> ORCH[Workflow Orchestrator]
    CONSENT[Consent Boundary] --> STORE[Cache and Manifest Store]

    UI --> ORCH
    UI --> LLM[Local LLM Engine]
    ORCH --> EVAL[Evaluation Engine]
    LLM --> EVAL
    EVAL --> TRAIN[Training Content Engine]

    TRAIN --> VOICE[XTTS Voice Generation]
    VOICE --> LIP[Lip-Sync Rendering]
    LIP --> VIDEO[Browser-Safe Video Layer]

    STORE --> TRAIN
    STORE --> VIDEO

    EVAL --> OUT[Candidate Report and Evidence]
    TRAIN --> OUT
    VIDEO --> OUT
    OUT --> DEPLOY[YouTube Demo + Hugging Face + GitHub + Wiki]
```

End-to-End Workflow

Stage	What Happens	Output
1. Setup	User provides resume, job description, topic, role, difficulty, and optional presenter/voice assets.	Structured candidate and training context
2. Question Generation	LLM generates resume-aware, JD-aware, technical, behavioral, and coding questions.	Interview question set
3. Candidate Capture	Candidate answers are captured with optional transcript and evidence metadata.	Candidate response record
4. Evaluation	Evaluation engine scores answer quality, clarity, role fit, confidence, and readiness.	Scorecard, strengths, weaknesses
5. Gap Analysis	Weak areas are converted into learning objectives.	Personalized training gaps

Stage	What Happens	Output
6. Training Plan	Training content engine creates modules, explanations, checkpoints, and scripts.	Adaptive training plan
7. Voice Generation	XTTS/Coqui generates speech from authorized voice or approved fallback.	Teaching audio file
8. Lip-Sync Rendering	SadTalker or Wav2Lip generates digital-human lecture video.	Rendered MP4
9. Browser-Safe Conversion	FFmpeg converts output to reliable browser playback format.	H.264/AAC MP4
10. Evidence Export	Reports, transcripts, scores, logs, and media manifest are packaged.	Evidence bundle

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Core Modules

1. Interview Intelligence

The interview engine uses the candidate resume and job description to produce targeted questions. Instead of generic prompts, it can focus on role expectations, project history, technical skills, behavioral readiness, and coding-style reasoning.

2. Evaluation Engine

The evaluation module scores responses and explains the reasoning behind the score. It can summarize strengths, identify weak areas, and convert gaps into actionable learning tasks.

3. Adaptive Training Planner

The training planner converts interview weaknesses into modules, explanations, checkpoint questions, and a teaching script. This makes the system useful not only for screening but also for improvement.

4. Digital Presenter Pipeline

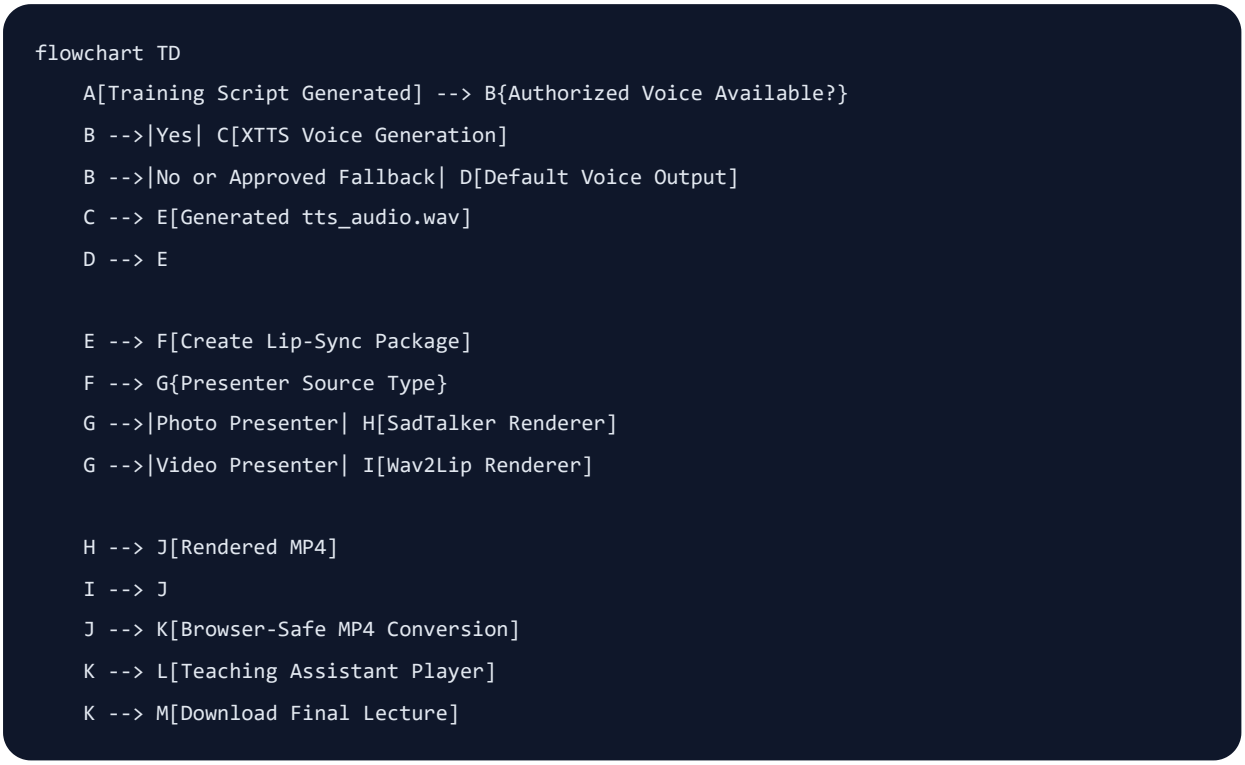
The digital presenter layer connects generated teaching scripts with voice synthesis and lip-sync rendering. This allows the system to create a teaching assistant experience rather than only producing text.

5. Evidence Pack Generator

The evidence layer makes the workflow auditable. It can export transcripts, evaluation summaries, JSON score records, training plans, generated audio/video manifests, and logs.

Digital Presenter Voice and Lip-Sync Pipeline

The digital presenter workflow is intentionally separated from the interview workflow. Candidate evidence should not automatically become presenter media. Presenter image/video and voice samples are separate assets and require explicit authorization.



> [!WARNING] Candidate face, voice, or identity should not be reused as digital presenter material unless the user has explicitly provided separate consent for that purpose.

Dashboard Experience

Dashboard Area	Reader Value
Interview Setup	Shows how resume, JD, role, and difficulty are configured.
Interview Session	Demonstrates question generation and response capture.
Coding Round	Supports technical or coding-style evaluation.
Evaluation Dashboard	Provides score, strengths, weaknesses, and readiness summary.
Training Hub	Converts gaps into personalized lessons and practice checkpoints.
Digital Presenter	Shows generated teaching script, voice options, presenter selection, and video preview.

Dashboard Area	Reader Value
Media Player	Plays generated browser-safe MP4 lectures.
Report Downloads	Allows export of candidate feedback, evidence, transcripts, and generated outputs.
Architecture Docs	Documents the complete workflow inside the app for reviewers.

Evidence Pack Structure

A strong AI system should not only produce an answer. It should also produce evidence. ASHU Mentor AI Studio is designed to package outputs for review, reproducibility, and demonstration.

```
report_exports/
  candidate_report_<timestamp>.md
  candidate_report_<timestamp>.json
  transcript_<timestamp>.txt
  interview_scores_<timestamp>.json
  training_plan_<timestamp>.md
  generated_teaching_script_<timestamp>.txt
  media_manifest_<timestamp>.json
  generated_audio/
    tts_audio.wav
  generated_video/
    final_lecture_browser_safe.mp4
  logs/
    app_log.txt
    voice_generation_log.txt
    lipsync_render_log.txt
```

The evidence package supports:

- candidate readiness review,
- technical interview coaching,
- learning-gap tracking,
- training recommendation,
- generated lecture replay,
- consent audit for presenter and voice assets,
- project demonstration through GitHub, Hugging Face, and YouTube.

Safety and Governance Model

Area	Safety Behavior
Candidate capture	Used only for interview evidence, evaluation, and reporting.
Presenter source	Must be separately authorized before digital-human rendering.
Voice sample	XTTS voice generation must use an authorized sample or approved fallback.
Candidate identity	Candidate face/voice is not reused as presenter unless separately authorized.
Fallback audio	Automated fallback voice requires user approval.
Generated media	Final lecture output is labeled as generated training content.
Reports	Candidate reports are evidence-backed and should not be treated as final hiring decisions without human review.
Deployment	Public demo surfaces should avoid private candidate data and unauthorized voice/presenter assets.

> **Candidate evidence is for evaluation. Presenter assets are for generated teaching. The two should never be mixed without explicit authorization.**

Runtime Example

```
# Clone the repository
git clone https://github.com/dranubhaparashar/ASHU-Mentor-AI-Studio.git
cd ASHU-Mentor-AI-Studio

# Activate the main app environment
conda activate chatbot

# Run the dashboard
ASHU_VOICE_ENV=voice \
COQUI_TOS_AGREED=1 \
PYTHONNOUSERSITE=1 \
python -m streamlit run app.py --server.fileWatcherType none
```

For voice environment validation:

```
conda activate voice

PYTHONNOUSERSITE=1 python -c "import numpy, scipy, torch; print('numpy', numpy.__version__);
print('scipy', scipy.__version__); print('torch', torch.__version__)"

COQUI_TOS_AGREED=1 PYTHONNOUSERSITE=1 python -c "from TTS.api import TTS;
TTS('tts_models/multilingual/multi-dataset/xtts_v2'); print('XTTS model load OK')"
```

Hugging Face Space Deployment

For Hugging Face deployment, keep the Space configuration at the top of `README.md` and keep `requirements.txt` as a clean package list.

```
---
title: ASHU Mentor AI Studio
emoji: 🎓
colorFrom: blue
colorTo: indigo
sdk: streamlit
sdk_version: "1.25.0"
app_file: app.py
pinned: false
---
```

The Hugging Face Space is useful for public UI demonstration, project showcasing, and live walkthroughs. Full local voice cloning, private file paths, and heavy lip-sync rendering may work better in local WSL/Conda environments.

Public Demo Surfaces

Surface	Purpose
Live Hugging Face App	Public browser-based demo of the Streamlit interface.
YouTube Demo	Walkthrough for reviewers, recruiters, collaborators, and readers.
dranubhaparashar / ASHU-Mentor-AI-Studio	Source code, app structure, technical assets, and deployment instructions.
GitHub Wiki Home	Architecture, technical specification, security governance, and user documentation.

Project Links

dranubhaparashar / ASHU-Mentor-AI-Studio

- **Live App:** ASHU Mentor AI Studio on Hugging Face Spaces
- **Demo Video:** ASHU Mentor AI Studio YouTube Demo
- **Wiki Home:** ASHU Mentor AI Studio Wiki
- **Demo & Live App:** Demo and deployment links
- **Architecture:** Architecture & Product Design
- **Technical Specification:** Complete Technical Specification
- **Training Workflow:** Training and Interview Workflow
- **Digital Presenter Pipeline:** Digital Presenter Voice and Lip-Sync Pipeline
- **Security and Consent:** Security, Consent and Governance

Documentation and Wiki Links

Wiki Page	Purpose
Home	Main entry point for the project wiki.
About ASHU Mentor AI Studio	Short overview, motivation, and project positioning.
Demo & Live App	YouTube demo, Hugging Face app, and public walkthrough links.
Architecture & Product Design	System design, runtime layers, and environment separation.
Complete Technical Specification	Full technical design and implementation reference.
Training & Interview Workflow	End-to-end interview, evaluation, and adaptive training flow.
Digital Presenter Voice & Lip-Sync Pipeline	Voice generation, presenter assets, and video rendering pipeline.
Background Rendering & Cache Reuse	Prefetching, cache reuse, and faster demo execution.
Feature Reference	Feature-by-feature explanation for readers and reviewers.
Acceptance Criteria Mapping	How implemented modules map to expected functionality.
Evidence Pack & Reporting	Reports, transcripts, scores, logs, and exported evidence bundle.
Security, Consent & Governance	Candidate evidence, voice authorization, presenter consent, and safety boundaries.
Quick Links	Fast navigation page for repository, wiki, demo, and app links.

Wiki Page	Purpose
GitHub Wiki Upload Instructions	Instructions for maintaining or uploading wiki pages.

Practical Use Cases

User Group	How ASHU Helps
Students	Practice interviews based on their own resume and target job.
Job Seekers	Identify role gaps and receive adaptive learning recommendations.
Interview Coaches	Use structured evidence to guide candidate improvement.
Educators	Convert topics into AI-generated teaching scripts and digital lectures.
HR Teams	Support pre-screening workflows with human review and evidence-backed reports.
AI Researchers	Explore combined LLM reasoning, evaluation, voice synthesis, lip-sync, and governance.
Product Demonstrators	Showcase an end-to-end GenAI workflow with app, demo, code, wiki, and video.

Engineering Value

ASHU Mentor AI Studio demonstrates how multiple AI capabilities can be connected into one practical system:

- **LLM orchestration** for interview and training generation,
- **evaluation logic** for answer scoring and readiness analysis,
- **adaptive training** for personalized learning paths,
- **voice generation** using XTTS/Coqui,
- **lip-sync rendering** using SadTalker or Wav2Lip,
- **browser-safe media conversion** using FFmpeg,
- **evidence export** for reports, transcripts, logs, and media manifests,
- **public demo readiness** through Hugging Face, YouTube, GitHub, and Wiki documentation.

It is designed to be measurable, explainable, adaptive, auditable, consent-aware, and demo-ready.

Current Strengths

- Complete interview-to-training workflow instead of isolated question generation.
- Resume-aware and JD-aware questioning for personalized interview preparation.
- Candidate response scoring with strengths, weaknesses, and readiness summary.
- Adaptive training plan generation from actual learning gaps.
- Digital-human lecture generation using voice and lip-sync pipelines.
- Cached media reuse for smoother demonstrations.
- Evidence pack export for review and reproducibility.
- Public project surface through GitHub, Hugging Face, YouTube, and Wiki.

Next Improvements

- Stabilize Hugging Face deployment for more reliable public access.
- Add role-specific rubrics for software, data science, AI, product, and management roles.
- Add exportable PDF reports for candidate feedback and training plans.
- Add structured coding-round scoring with test cases and explanations.
- Add a consent checklist UI before voice or presenter rendering.
- Add model-provider abstraction for local LLMs, OpenAI, Azure OpenAI, and Gemini.
- Add dashboard-level evidence pack export for demo and review workflows.
- Add transcript alignment for generated lecture videos.
- Add analytics for comparing candidate progress across multiple interview attempts.

Key Innovation

> [!IMPORTANT] The strongest innovation is not one model or one screen. The innovation is the complete connected pipeline: **resume-aware interview → JD-aware evaluation → learning-gap analysis → adaptive training → voice generation → lip-sync rendering → browser-safe video → evidence-backed report**.

Most tools handle these steps separately. ASHU Mentor AI Studio connects them into one workflow so that the reader can see a complete AI product idea, not only a prototype screen.

Conclusion

ASHU Mentor AI Studio shows how interview preparation and training can evolve from disconnected tools into an end-to-end AI mentoring platform.

It combines structured interview intelligence, explainable evaluation, personalized learning, consent-aware media generation, and public demo documentation into one integrated system.

The result is a practical AI mentor studio for users who need more than mock questions. It supports assessment, improvement, teaching, demonstration, and evidence-backed review in one workflow.

Final Thought

> From **interview practice** to **adaptive digital-human training**.
>
> The real value is not only asking better questions. It is connecting questions, answers, evaluation, learning gaps, teaching scripts, generated voice, rendered video, and evidence-backed reports into one explainable AI workflow.
>
> **Resume-aware interview · JD-aware evaluation · candidate evidence · adaptive training · consent-based voice · digital-human lecture · GitHub + Hugging Face + YouTube demo surface**

Source: src/content/posts/ASHU Mentor AI Studio/ashu-mentor-ai-studio.md